

Mechanism Black Box Student Handout

TOK Cognitive Science Activity Bank • Natural Sciences • 30-40 min

Category	Natural Sciences	Time	30-40 min
Difficulty	Medium	ID	mechanism-black-box

Big idea: Scientific explanation often goes beyond prediction by proposing mechanisms.

Student-Facing Prompt

Inspect Mechanism Black Box Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Mechanism Black Box Stimulus A	Student-facing stimulus 1: Rule box: input a number; output is double plus three. Competing mechanisms may fit early examples. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Mechanism Black Box Stimulus B	Student-facing stimulus 2: Physical box: different objects go in; the teacher secretly applies a sorting rule before returning an output card. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Use this for comparison, a second round, or a partner challenge.
Mechanism Black Box Reveal / Twist	Student-facing stimulus 3: Physical box: different objects go in; the teacher secretly applies a sorting rule before returning an output card. Identify the observation, variable, control, anomaly, or missing method detail before making a claim.	Teacher reveal: connect the changed response to the big idea — Scientific explanation often goes beyond prediction by proposing mechanisms.

Actual Lesson Questions

#	Question for students
1	After inspecting Mechanism Black Box Stimulus A, what is your first knowledge claim?
2	Which exact detail from Mechanism Black Box Stimulus A did you use as evidence?
3	What did you infer beyond the evidence?
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?
5	What missing information would most improve the knowledge claim?
6	What is added by knowing a mechanism?

Ready-to-Use Examples

- Rule box: input a number; output is double plus three. Competing mechanisms may fit early examples.
- Physical box: different objects go in; the teacher secretly applies a sorting rule before returning an output card.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Natural Sciences.

Activity Steps

1. Introduce a black box that produces outputs from inputs according to a hidden rule.
2. Groups test inputs and record outputs.
3. Students propose a mechanism that could explain the pattern.
4. Groups test a new input to discriminate between mechanisms.
5. Debrief why explanation requires more than curve-fitting.

Observation and Evidence Record

Use this grid to keep observation, inference, evidence, and confidence separate.

What I noticed	What I inferred	Evidence	Confidence

TOK Debrief

- Knowledge question: What is added by knowing a mechanism?
- Can a model predict well without explaining?
- How do scientists choose between competing unseen explanations?

Knowledge Questions

- What is added by knowing a mechanism?
- Can prediction count as explanation?
- How is unseen scientific knowledge justified?

Transfer Example

For example, run this activity with a system with inputs and outputs but hidden mechanism. Have students commit to an initial claim, then reveal the change, hidden rule, alternative context, or comparison that makes the knowledge problem visible.

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...